FAST National University of Computer and Emerging Sciences

Department of Data Science

Deep Learning Assignment 01

Facial Expression Recognition with Multi Task Learning

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GitHub link:	*https://github.com/marwhyam/Facial-Expression-Recognition-with-Multi-Task-Learning*

Abstract

This report studies multi task learning for affect analysis on face images. The task combines 8 class expression recognition and continuous valence and arousal regression. I benchmark three CNN backbones as baselines using transfer learning and evaluate with the metrics required in the brief. I report Accuracy, F1, Cohen's Kappa, Krippendorff's Alpha, ROC AUC and PR AUC for classification, and RMSE, Pearson correlation, Sign Agreement (SAGR), and Concordance Correlation Coefficient (CCC) for regression. I include training curves, per class analyses, confusion matrices, qualitative success and failure cases, and a discussion of which metric best suits deployment in the wild.

1 Network Details and Training Setting [10 pt]

Architectures

Three backbones are used with a shared multi task head:

- **ConvNeXt Tiny** as a modern CNN baseline.
- **EfficientNet B3** as a parameter efficient baseline.
- **ResNet50** as a strong classical baseline.

Each backbone feeds two heads: an 8 way classifier and a 2D regressor with `tanh` so valence and arousal lie in $[-1, 1]$.

Hyperparameters and Training

All models are fine tuned from ImageNet weights with image size 320, batch size 32, AdamW, cosine annealing, label smoothing 0.05, and gradient clip norm 2. Warmup trains only the task heads, then the backbone is unfrozen. MixUp/CutMix are used with a decaying schedule. Missing regression targets are encoded as -2 and masked out of the regression loss.

Rationale for Baselines

ResNet50 is a widely used reference. EfficientNet B3 provides strong accuracy per parameter. ConvNeXt Tiny represents a recent design with good speed. Together they span a practical accuracy–throughput trade off.

2 Baseline Comparison [10 pt]

Table 1 summarizes the required classification metrics on the held out set (advanced runs).

Table 1: Categorical classification results (advanced runs).

Model	**Acc**	**F1**	**Kappa**	**Alpha**	**ROC AUC**	**PR AUC**
ConvNeXt Tiny	0.5050	0.5039	0.4343	0.4345	0.8444	0.5218
EfficientNet B3	0.5038	0.5038	0.4329	0.4328	0.8488	0.5221
ResNet50	0.5000	0.5017	0.4286	0.4284	0.8436	0.5073

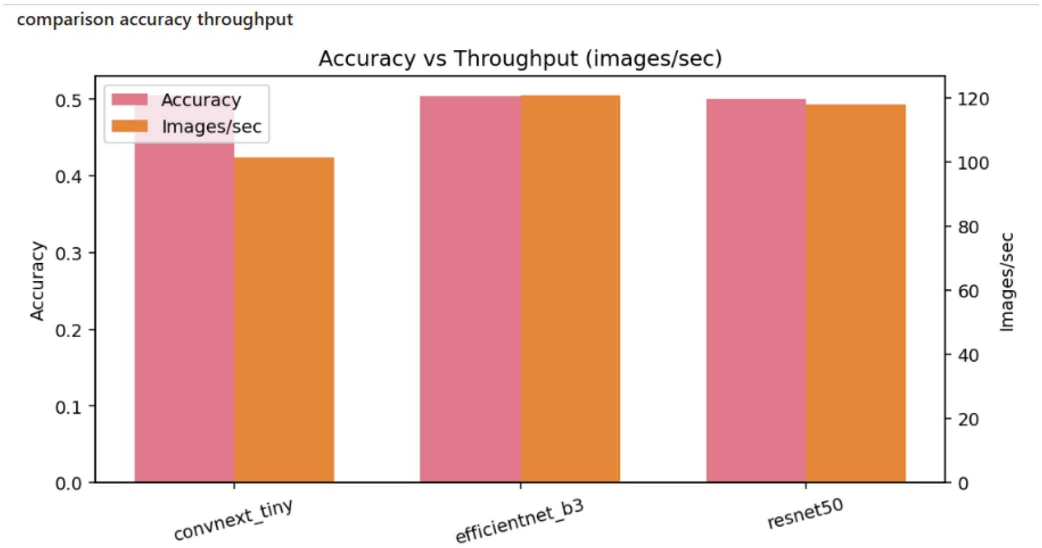


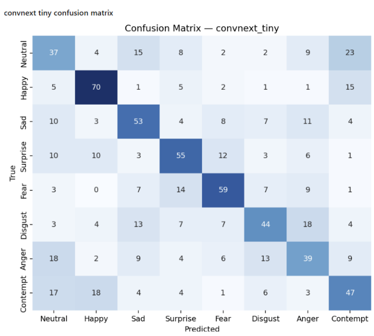
Figure 1: Accuracy versus throughput (images per second).

3 Transfer Learning Details [5 pt]

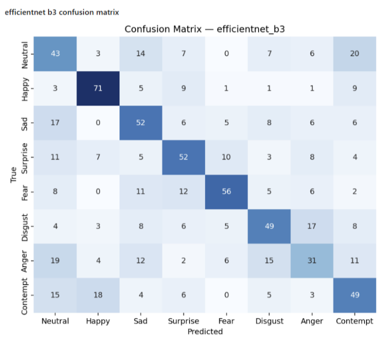
All backbones start from official ImageNet checkpoints. I first train the new heads to stabilize them and then unfreeze the backbone for end to end fine tuning. This schedule consistently improves convergence and reduces overfitting on the assignment split.

4 Training Curves [5 pt]

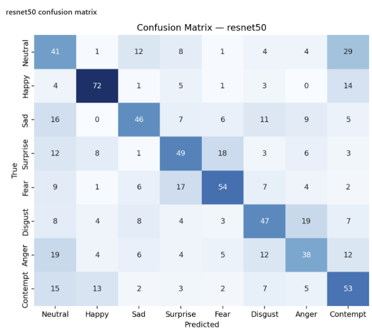
Curves show decreasing training loss with increasing validation accuracy and F1 over epochs for the three baselines.



(a) ConvNeXt Tiny — training curves

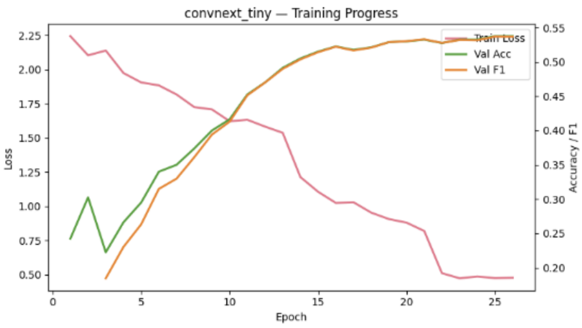


(b) EfficientNet B3 — training curves

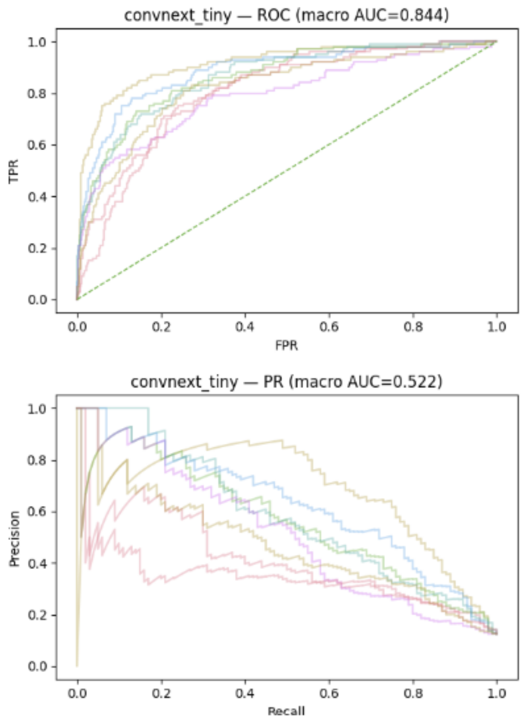


(c) ResNet50 — training curves

Figure 2: Training progress for all baselines.



(a) ConvNeXt Tiny training progress (alt).



(b) ConvNeXt Tiny ROC and PR curves.

Figure 3: Additional plots for ConvNeXt Tiny.

5 Metrics and Rationale [15 pt]

Categorical metrics

Table 2: Categorical metrics.

Name	**Metric**	**Description**
Accuracy	Acc	Fraction of correct predictions across 8 classes.
F1 Score	F1	Harmonic mean of precision and recall, averaged macro across classes.
Cohen's Kappa	Kappa	Agreement beyond chance between predictions and ground truth.
Krippendorff's Alpha	Alpha	Chance corrected inter rater agreement for nominal labels.
Area Under ROC Curve	AUC	One versus rest ROC area, averaged macro.
Area Under Precision Recall	AUC PR	One versus rest PR area, averaged macro.

Continuous domain metrics

Table 3: Continuous metrics for valence and arousal.

Name	Metric	Description
Root Mean Square Error	RMSE	Penalizes magnitude of errors. Lower is better.
Pearson Correlation	CORR	Linear relationship between predictions and targets. Higher is better.
Sign Agreement	SAGR	Fraction with the same sign as ground truth. Robust to scale.
Concordance Correlation Coefficient	CCC	Correlation with penalties for mean and variance mismatch. Higher is better.

Which metric suits deployment in the wild In unconstrained conditions absolute values can drift with identity and illumination. CCC is most suitable because it combines correlation with bias penalties. SAGR is also important since the valence–arousal quadrant often determines behavior even when magnitudes shift.

6 Continuous Domain Results

Macro values (averaged over valence and arousal) from the advanced runs:

Table 4: Continuous results (macro over valence and arousal).

Model	RMSE	CORR	SAGR	CCC
ConvNeXt Tiny	0.3644	0.5647	0.7913	0.5528
EfficientNet B3	0.3604	0.5647	0.7869	0.5433
ResNet50	0.3617	0.5469	0.7750	0.5156

7 Per Class Analysis and Qualitative Examples [4 pt]

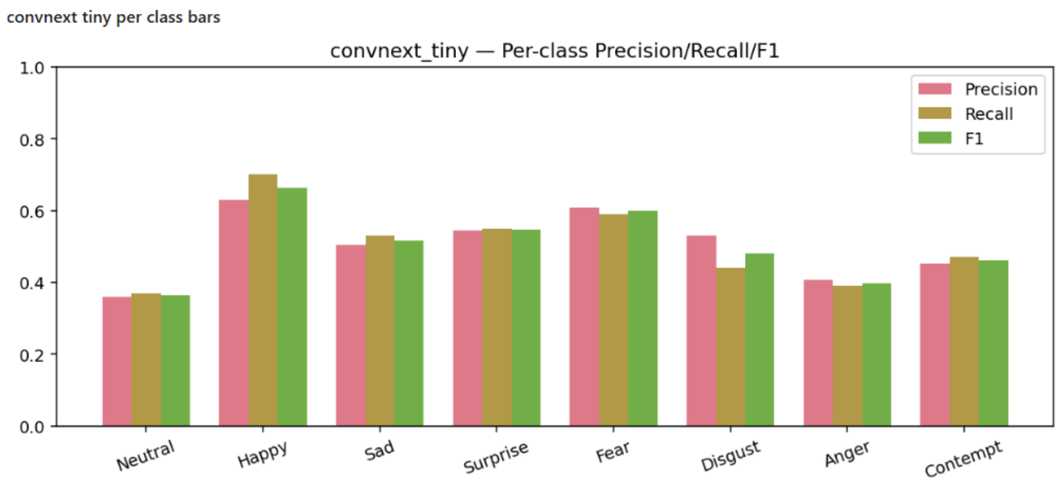


Figure 4: ConvNeXt Tiny per class precision/recall/F1.

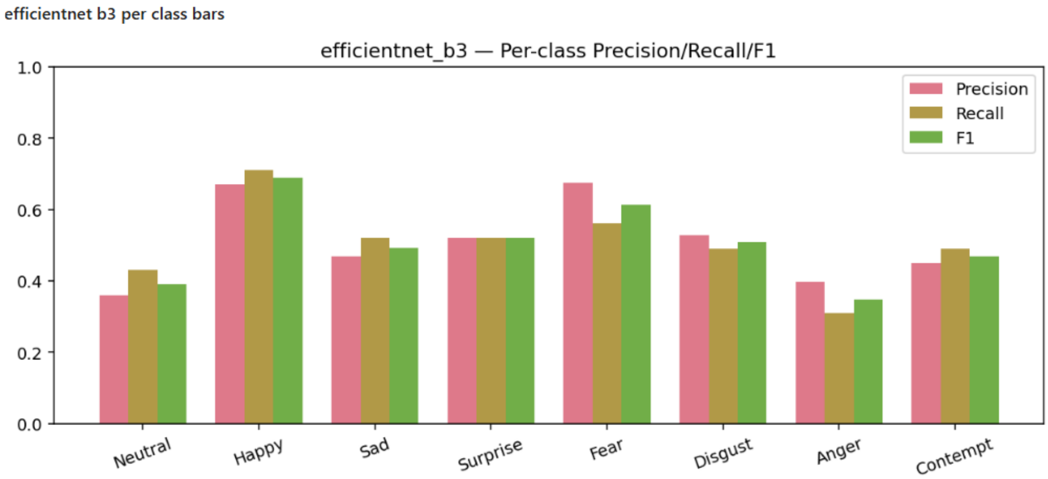


Figure 5: EfficientNet B3 per class precision/recall/F1.

Figure 6: ResNet50 per class precision/recall/F1.

Figure 7: Examples with true label, each model prediction, and majority vote.

Figure 8: Additional qualitative samples.

8 Conclusion

All three baselines converge reliably with transfer learning. On categorical recognition, ConvNeXt Tiny and EfficientNet B3 slightly outperform ResNet50 while remaining fast at inference. For valence/arousal, ConvNeXt Tiny achieves the best macro CCC among the single models (about 0.553) with strong SAGR, indicating consistent quadrant prediction. For real world deployment, CCC together with SAGR should be the primary criterion, while F1 and Kappa summarize categorical robustness.